HDOC Day-66: Recursive Matrix Multiplication in C

1. Aim

To write a C program to multiply two square matrices using recursion without using iterative loops for the multiplication logic.

2. Problem Statement

Write a recursive program to multiply two square matrices of size N × N. The multiplication must be performed using recursion instead of for or while loops.

3. Algorithm

1. Start the program.
2. Read the order n of the square matrices.
3. Input the elements of matrices A and B.
4. Initialize the resultant matrix C with zeros.
5. Define a recursive function multiply() using three indices i, j, and k.
6. Multiply A[i][k] with B[k][j] and add the result to C[i][j].
7. Recursively increase k to calculate each element.
8. When k == n, move to the next column by increasing j.
9. When j == n, move to the next row by increasing i.
10. Stop recursion when i == n.
11. Display the resultant matrix.
12. Stop the program.

4. C Program

```
#include <stdio.h>

void multiply(int n, int A[n][n], int B[n][n],
              int C[n][n], int i, int j, int k)
{
   if (i == n)
      return;

   if (j == n)
   {
```

```
        multiply(n, A, B, C, i + 1, 0, 0);
        return;
    }

    if (k == n)
    {
        multiply(n, A, B, C, i, j + 1, 0);
        return;
    }

    C[i][j] += A[i][k] * B[k][j];

    multiply(n, A, B, C, i, j, k + 1);
}

int main()
{
    int n;

    printf("Enter order of matrices: ");
    scanf("%d", &n);

    if (n <= 0 || n > 10)
    {
        printf("Invalid matrix size");
        return 1;
    }

    int A[n][n], B[n][n], C[n][n];

    printf("Enter first matrix:\n");
```

```
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &A[i][j]);

    printf("Enter second matrix:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &B[i][j]);

    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            C[i][j] = 0;

    multiply(n, A, B, C, 0, 0, 0);

    printf("Resultant Matrix:\n");
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
            printf("%d ", C[i][j]);

        printf("\n");
    }

    return 0;
}
```